

RED SEAL EXAM PREP

308A Sheet Metal Worker

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
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Mock Exam — answer all questions, then check the key

Q1. A sheet metal worker is applying flashing near an unguarded roof edge. How does a travel restraint system differ from a fall arrest system?

- A) Restraint requires a full harness; arrest allows a belt
- B) Restraint prevents reaching the edge; arrest stops a fall
- C) Restraint suits sloped roofs; arrest suits only flat ones
- D) Restraint needs no anchor; arrest needs a rated anchor

Q2. When developing the pattern for a right cone using radial line development, what measurement represents the radius of the arc drawn in the pattern?

- A) The diameter of the cone base
- B) The radius of the cone base
- C) The vertical height of the cone
- D) The slant height of the cone

Q3. A length of round duct is cut off on a slant at one end. Seen square on, that cut opening is an ellipse. The piece is now developed by the parallel line method: the stretch-out is stepped off in equal divisions, and the height of the cut at each division is projected across from the elevation. What shape do the plotted heights trace along the cut edge of the pattern?

- A) A straight line, since the cut is made on a flat plane
- B) An ellipse, matching the shape of the cut opening
- C) A sine wave, one full cycle across the stretch-out
- D) A circular arc struck from a single centre point

Q4. In SMACNA's notes for the transverse joints of Figure 1-4, one of these joints is held to a 2 in. w.g. maximum static pressure. Which one?

- A) Welded flange
- B) Companion angle flange
- C) Standing S slip
- D) Flat S slip

Q5. What is the squaring arm on a squaring shear used for?

- A) Setting the distance the sheet is pushed past the blade
- B) Holding the sheet square to the blade before the cut
- C) Setting the running clearance between upper and lower blades
- D) Holding the sheet down on the table while the blade cuts

Q6. What is "springback" in sheet metal bending and how is it compensated?

- A) Stretching at the bend — corrected by adding material
- B) Thermal expansion during welding — corrected by pre-heating
- C) Elastic return after bending — corrected by overbending
- D) Sheet vibration during cutting — corrected by clamping

Q7. A concentric round-to-round reducer is a frustum, a right cone with its top cut off. Using radial line development, what must be established before any pattern arc can be struck?

- A) The seam allowance to be added at the shortest element
- B) The true length of each diagonal across the tapered face
- C) The apex, found where the slant sides would meet
- D) The stretch-out, set equal to the small end circumference

Q8. A rectangular duct is to be built to the SMACNA rectangular duct reinforcement schedule. Which factors together fix the minimum galvanized steel gauge of the duct wall?

- A) Duct size by itself, whatever the pressure class
- B) Pressure class and duct size, taken by themselves
- C) Air velocity and the overall length of the duct run
- D) Pressure class, size and reinforcement spacing

Q9. What is the purpose of a turning vane installed inside a 90° rectangular duct elbow?

- A) To guide airflow smoothly around the bend, reducing turbulence and noise
- B) To stiffen the elbow corners so a lighter gauge of sheet can be used at the bend
- C) To split the airflow evenly between two branches taking off at the elbow, balancing the system
- D) To break the airstream into layers so condensation cannot form on the elbow's inner wall

Q10. When laying out a round elbow using parallel line development, why is the stretchout line equal to the circumference of the duct?

- A) To account for material stretch during bending
- B) SMACNA mandates this for pressure class compliance
- C) Because the stretchout is the unrolled cylinder surface (πd)
- D) It is equal to the diameter only, not the circumference

Q11. A VAV (Variable Air Volume) system modulates airflow to individual zones. What controls the airflow at each zone terminal?

- A) A VAV box with a thermostat-controlled damper
- B) Manual balancing dampers set at commissioning
- C) A fixed orifice plate in the branch duct
- D) The AHU supply fan speed alone

Q12. Two folded edges on a round duct have been hooked together to form a grooved seam. Which hand tool locks that seam?

- A) A hand crimper worked along the seam
- B) A hand groover sized to the seam width
- C) A hand seamer, which folds a flat edge
- D) A setting hammer worked along the seam

Q13. An oxy-acetylene outfit is being hooked up for cutting work in the shop. Under the BC Occupational Health and Safety Regulation, where must the safety devices that prevent reverse gas flow and arrest a flashback be installed?

- A) On piped gas lines, but not on a portable cart outfit
- B) At each cylinder valve, ahead of its own regulator
- C) On each hose, in the run between regulator and torch
- D) On the fuel gas hose only, since it feeds the flame

Q14. According to SMACNA, pressure class for HVAC ductwork is defined by the maximum static pressure the duct system will operate under. Which of the following correctly lists SMACNA low-pressure classes?

- A) Class A, B, C based on material gauge only
- B) Class 100, 200, 300 based on CFM airflow
- C) Class 1/2, Class 1, Class 2 (in inches of water gauge, WG)
- D) Pressure class is not defined by SMACNA — it is determined by the HVAC engineer only

Q15. What is the primary purpose of insulating HVAC supply ductwork in an unconditioned attic?

- A) To meet SMACNA pressure class requirements
- B) To prevent condensation and reduce thermal loss
- C) To reduce duct velocity noise in the attic
- D) To allow higher static pressure ratings

Q16. When working in a ceiling plenum space above a suspended ceiling, what is a key safety concern specific to sheet metal workers?

- A) Chemical burns from duct sealant curing in the enclosed space
- B) Sustained noise above 85 dBA from the operating HVAC equipment
- C) Energised conduit contact and falls through the grid
- D) UV exposure from the fluorescent and LED fixtures overhead

Q17. A sheet metal worker is installing a flexible connection (flex connector) between the AHU and rigid ductwork. What is the primary purpose?

- A) To isolate AHU fan vibration from the ductwork
- B) To act as the system's volume control damper at the unit
- C) To increase the cross-sectional area leaving the unit
- D) To allow the duct to be taken apart quickly for cleaning access

Q18. Heavy blue-black heat tint is found on a TIG welded stainless steel duct seam that has to keep its corrosion resistance. What removes it correctly?

- A) Pickling, or abrasives kept for stainless only
- B) A carbon steel wire brush and a solvent wipe
- C) Leaving it, since the oxide film protects the steel
- D) Reheating the seam until the tint burns off

Q19. A sheet metal worker is TIG welding a 16-gauge (1.6 mm) 304 stainless steel duct section. The band around the finished weld comes out dark blue and grey. What does this indicate and what is the cause?

- A) Mill scale, so the sheet is carbon steel and not 304
- B) Ordinary weld colouring on stainless, needing no action
- C) The filler rod composition is wrong for 304 stainless
- D) Heavy heat tint, from lost shielding gas coverage

Q20. Under BC and most Canadian OH&S regulations, what is the maximum height from which a worker can fall before fall protection is mandatory on a construction site?

- A) 6m (20 ft)
- B) 1.5m (5 ft)
- C) 3m (10 ft)
- D) 4.5m (15 ft)

Q21. When using a slip roll to form a cone (tapered cylinder), how is the taper achieved?

- A) A tapered forming die is inserted between the rolls
- B) One end of the front pinch rolls is set tighter than the other
- C) The sheet is fed at an angle using a guide fence
- D) One end of the rear roller is raised

Q22. Which pattern development method is used to lay out a straight rectangular duct section?

- A) Development by triangulation
- B) Radial line development
- C) Parallel line development
- D) Gore segment construction

Q23. A shop is making slip-and-drive transverse joints on rectangular duct that has to be sealed to a specified sealing class. SMACNA warns that two things about the connection must be controlled or the sealing will be compromised. What are they?

- A) Connector length and notch depth at the duct ends
- B) Sheet gauge and the pressure class of the section
- C) Rivet spacing and the size of the rivet head used
- D) Sealant brand and the temperature at application

Q24. A grooved seam is commonly used in which application?

- A) As the spiral seam on round duct and on joined curved edges
- B) As a transverse (cross) connection between rectangular duct sections
- C) As a standing seam on architectural metal roofing
- D) As the longitudinal seam on rectangular duct sections

Q25. In British Columbia, a sheet metal worker is cutting on a layout table when a sheet slips and lacerates his forearm. He is treated at a walk-in clinic, where the wound is closed with sutures, and he returns to finish his shift. Under the Workers Compensation Act, the employer must:

- A) Investigate it, but the clinic reports the injury
- B) Report the injury and investigate the incident
- C) Report the injury; no investigation is required
- D) Record it as first aid, since no time was lost

Q26. What type of snips is designed to cut straight lines and large curves in sheet metal?

- A) Yellow-handled aviation snips (straight pattern)
- B) Red-handled aviation snips (left-cut pattern)
- C) Hawk's-bill snips (tight-radius circle pattern)
- D) Green-handled aviation snips (right-cut pattern)

Q27. A rectangular duct section 400 mm by 250 mm is to be made from one sheet, closed with a Pittsburgh lock at one corner. What width of flat blank does the layout call for?

- A) 1300 mm, plus the pocket and edge allowances
- B) 1300 mm, the girth already covering the lock
- C) 1300 mm, with the lock allowance taken out of it
- D) 650 mm, plus the pocket and edge allowances

Q28. What is the purpose of tack welds during sheet metal assembly?

- A) To provide the final weld strength alone
- B) To burn mill scale off the joint edges
- C) To test electrode compatibility first
- D) To hold parts in position before welding

Q29. A panel is being run through a Pittsburgh lock former to form the pocket on one edge. How is the panel presented to the machine?

- A) Standing on edge in the rolls; panel orientation makes no difference
- B) Lying flat, the edge fed squarely into the rolls against the guide
- C) Lying flat, but entered at about 45 degrees to the line of the rolls
- D) Entered from the outfeed side so the pocket runs toward the operator

Q30. A sheet metal worker is developing a pattern for an offset transition fitting — a rectangular duct offset to one side connecting to a round duct. Which sequence correctly describes the triangulation method?

- A) Divide only the round end into segments → project them directly to the square end → connect points
- B) Find the apex of the cone → draw radial lines from the apex → swing arcs for each section
- C) Draw plan and elevation → divide surfaces into triangles → find true lengths → build the pattern
- D) Draw the elevation only → project parallel lines to a stretch-out line → mark off the perimeter

Q31. What is the purpose of a balancing damper installed in a duct branch?

- A) To prevent backflow in the branch duct
- B) To throttle branch airflow to the design CFM
- C) To maintain constant static pressure in the main
- D) To open automatically on a call for cooling

Q32. A pattern is being laid out for a piece that will be finished with a wired edge. What allowance is added along that edge, and what is it based on?

- A) A fixed 13 mm, whatever size wire is used
- B) One wire diameter plus one sheet thickness
- C) About two and a half times the sheet thickness
- D) About two and a half times the wire diameter

Q33. Which shop machine forms the narrow flanged "male" edge that enters the pocket of a Pittsburgh lock?

- A) Pittsburgh lock former
- B) Hand notcher
- C) Bar folder
- D) Slip roll former

Q34. A box and pan brake is used for what sheet metal forming operation?

- A) Forming the Pittsburgh lock seam on duct sections and corners
- B) Bending sheet metal into boxes and pans at precise angles
- C) Rolling flat sheet into cylinders and round duct sections
- D) Cutting sheet metal blanks to exact length before any forming

Q35. Before starting any work on an energised electrical panel in the mechanical room where ductwork is being installed, the sheet metal worker must:

- A) Wear rubber insulating gloves only
- B) Simply turn off the panel breaker and proceed
- C) Apply LOTO procedures on all energy sources
- D) Notify the foreman verbally and proceed

Q36. When developing a pattern for a cylindrical elbow using parallel line development, what determines the number of equal divisions the circumference is broken into?

- A) The gauge of the sheet being formed and its bend allowance
- B) The accuracy required in the finished pattern curve
- C) The centreline radius of the elbow divided by the duct diameter
- D) The number of gores or pieces making up the elbow

Q37. What is a Pittsburgh lock seam and for what application is it primarily used?

- A) A double-folded flat seam used on architectural metal roofing
- B) A longitudinal seam used to close the sides of rectangular duct sections
- C) A cross-seam used to connect two sections of rectangular duct end-to-end
- D) A spiral seam used on round ductwork formed by a spiral pipe machine

Q38. A 90 degree round elbow is to be made in four pieces, laid out so that every mitre cut in the fitting is the same. Measured from a line square across the pipe, what is the angle of each mitre cut?

- A) 45 degrees
- B) 22.5 degrees
- C) 30 degrees
- D) 15 degrees

Q39. ASHRAE 62.1 governs ventilation for acceptable indoor air quality. Which parameter does it specify for commercial occupancies?

- A) Minimum outdoor air rate per person and floor area
- B) Maximum permitted duct velocity in metres per second
- C) Minimum refrigerant charge weight per equipment room
- D) Maximum allowable CO₂ concentration in the flue gas

Q40. A sheet metal worker is asked to install a transfer air grille between two rooms to maintain pressure balance. Where should it be located?

- A) Location does not matter — only the grille size matters
- B) At high level (near ceiling) in the partition wall
- C) At mid-height in the neutral pressure zone
- D) At low level near the floor for cold air transfer

Q41. When cutting sheet metal with aviation snips, where should the waste side of the material be kept?

- A) On the side the ramped jaw faces, so the waste lifts clear
- B) Either side; the snip colour only sets which way a cut can curve
- C) Under the lower blade, so the waste curls down clear of the handles
- D) On the left in every case, as snips are built for right-handers

Q42. When soldering a flat lock seam (folded seam) on tinplate, what is the correct heating technique to ensure full solder penetration?

- A) Heat the seam itself until it melts solder on contact
- B) Melt bulk solder over the seam with a torch and let it flow
- C) Apply solder to the iron tip, then wipe across the cold seam
- D) Heat the solder wire directly until it melts into the seam

Q43. Brazing differs from welding in that:

- A) Brazing is used only for aluminium work
- B) The filler melts but the base metal does not
- C) Brazing melts the base metal; welding does not
- D) Brazing requires flux; welding never does

Q44. A round-to-round eccentric reducer is to be developed. Its apex is not centred over the two openings. Which statement is correct?

- A) Radial line development applies directly, as it is still a right cone
- B) Each element line must be found in true length, as they differ
- C) Parallel line development applies, since both openings are round
- D) A concentric reducer pattern will do if the small end is cut on a slant

Q45. What does the acronym AHU stand for in HVAC systems?

- A) Auxiliary Heating Unit
- B) Airflow Humidity Unit
- C) Automatic Heat Updraft
- D) Air Handling Unit

Q46. A rooftop exhaust fan is shut down overnight and outdoor air blows back down the duct into the building. Which unpowered damper is fitted in the duct to stop this?

- A) A motorised isolation damper
- B) A fire damper with a fusible link
- C) A backdraft (gravity) damper
- D) A manual volume control damper

Q47. Resistance spot welding is commonly used in sheet metal fabrication because it:

- A) Can join galvanized steel to aluminum without special preparation
- B) Joins sheets at discrete spots using localized resistance heating
- C) Requires flux to prevent oxidation at each weld spot
- D) Creates the strongest possible joint for all metal thicknesses

Q48. Sheet metal edges left from snipping or shearing are extremely sharp. The correct PPE to handle cut sheet metal is:

- A) Standard leather work gloves
- B) Rubber chemical gloves
- C) Cut-resistant (level A4 or higher) gloves
- D) No gloves — gloves catch on machinery

Q49. A "rotary machine" (power rotary) in a sheet metal shop can form all of the following EXCEPT:

- A) Beads and swages rolled into a panel
- B) Turned edges and flanges
- C) Punched rivet holes
- D) Wired edges

Q50. A duct system shows higher than design static pressure drop across a filter bank. What is the most likely cause?

- A) Filters loaded with dust and dirt
- B) Excessive outdoor air intake volume
- C) Supply fan speed set too low
- D) Duct leakage downstream of the filter

Answer Key & Explanations

Q1. Answer: B

The two systems answer different questions. A travel restraint system is rigged so the worker physically cannot get to the edge: the lanyard is short enough that the fall never happens. A fall arrest system assumes the worker will go over and is built to stop the fall before they strike anything below. That is why the regulations rank restraint ahead of arrest. The Ontario construction regulation requires a guardrail system first, and where that is not practicable, the highest ranked practicable method from travel restraint, then a fall restricting system, then fall arrest, then a safety net. Roof slope does not decide which system is used. Both must be attached to a fixed support, so a restraint lanyard is anchored just as an arrest lanyard is. The hardware difference runs opposite to the common belief: travel restraint may use a full body harness or a safety belt, while fall arrest requires a full body harness with a shock-absorbing lanyard, because arresting a fall in a belt injures the worker.

Q2. Answer: D

In radial line development of a cone, all surface lines radiate from the apex. The radius of the arc in the flat pattern equals the slant height — the true length from the apex to the edge of the base circle. The arc length then equals the circumference of the base ($2\pi r$). The base radius fixes the included angle of that arc, not its radius, and the vertical height alone is always shorter than the slant height.

Q3. Answer: C

Two different curves are in play and the item turns on keeping them apart. The cut face, the opening seen square on, is a plane section of a cylinder and is an ellipse. The cut edge on the flat pattern is a different curve, because developing the piece lays the circumference out straight while leaving every height as it was. Why it comes out as a wave is visible in the elevation. The mitre line there is straight, and each height is read where that straight line crosses an element. But the elements do not project evenly onto the elevation: they crowd together toward the two outside edges of the pipe and open out through the middle, so a point standing at an angle θ around the pipe projects to a horizontal position proportional to the cosine of θ , and the straight mitre line therefore hands it a height proportional to the cosine of θ as well. On the stretch-out those same points are spaced evenly, one equal division apart. Replotting cosine heights at even spacings is exactly what turns the straight line into a wave: measured from the mid-height of the cut, the height is the radius times the tangent of the cut angle times the cosine of the angle travelled around the pipe, plotted against arc length. That is a sinusoid, and it runs one full cycle over the stretch-out, with a crest at the longest element, the mid-height crossed at the two side elements, and a trough at the shortest element. Whether it reads as a sine or a cosine depends only on which element the stretch-out is started from; the curve is the same. An ellipse answers a different question. It is the true shape of the hole, and it is what the cut looks like while the metal is still round. Setting an ellipse out on the blank instead gives a piece whose end will not lie in a plane once it is rolled up. No circular arc will fit the curve, whatever radius is chosen, because an arc bends the same way from end to end. This curve does not: it arches over the longest element and hollows under the shortest, reversing at the two side elements. That is why the layout is finished by joining the plotted points with a flexible curve rather than by swinging an arc from a centre. A straight line would be right for a prism, not a cylinder. Rectangular duct cut on a slant does develop with straight pattern edges, one straight run across each face, and that is where the expectation comes from. On round duct the mitre line is straight only in the elevation, where the elements are foreshortened; once the even stretch-out spacings replace those foreshortened ones, the line becomes a wave.

Q4. Answer: D

SMACNA's notes for Figure 1-4 give joint T-5, T-6, flat S slips, a line of their own: the slip is 24 gauge minimum and no lighter than two gauges below the duct, and when it is used on all four sides it is fastened within 2 in. of the corners and at 12 in. maximum intervals, at 2 in. w.g. (500 Pa) maximum static pressure. No other joint offered here is held that low. The standing S connectors, joints T-10 through T-12, are worked to 4 in. w.g. and the same notes say they are not recommended above that. The welded flange, joint T-21, is given no pressure ceiling at all; its note deals with flange width, edge welding and corner reinforcement. The companion angle, joint T-22, is plainly meant to go higher, because its bolt schedule sets 5/16 in. minimum bolts at 6 in. maximum spacing on the 4 in. w.g. class or lower and 4 in. maximum spacing on higher pressures, and a joint would not need a rule for higher pressures if it were never used there. Read all of this as selection limits from a construction standard rather than as Canadian law: SMACNA governs where a project specification or a provincial code adopts it.

Q5. Answer: B

The squaring arm is the graduated arm that projects from the front of a squaring shear at ninety degrees to the blade. The sheet is pushed up against it so the edge being fed is square to the cut, which is what makes successive cuts on a blank true to one another, and the scale along the arm then lets the operator gauge the width of a piece taken from the front without measuring each one. It positions the work; it does not clamp it. Clamping is the hold-down, a bar or set of feet that comes down on the sheet just ahead of the blade an instant before the cut so the sheet cannot creep or lift. Running clearance between the upper and lower blades is a machine setting made at the blade mounts according to material thickness, and it is what decides whether the cut edge comes off clean, rolled over or chipped. Setting how far the sheet travels past the blade is the job of the back gauge, the stop behind the blade. That is a different device from the squaring arm, and it is the same device a press brake carries to repeat a flange width.

Q6. Answer: C

When a bend is released, elastic stress partially straightens the metal. To achieve the required final angle, the sheet must be overbent slightly. The overbend angle depends on material and thickness.

Q7. Answer: C

Radial line development swings every arc from one centre, the apex of the cone. A frustum has had that apex cut away, so it is recovered on paper by extending the two slant sides of the elevation until they intersect. The flat pattern is then the area between two arcs struck from that point: the outer radius is the true slant length from the apex to the large end, the inner radius the true slant length to the small end, and the included angle is set by stepping the large-end stretch-out around the outer arc. Without the apex there is no centre to swing from, which is why finding it is the first step and not the last. The stretch-out is stepped around the arc rather than replacing the apex, and it is taken from the large end, not the small one. Seam allowance is added after the true pattern is drawn. True lengths of diagonals belong to triangulation; a right cone does not need them, because every element line from the apex is the same slant length.

Q8. Answer: D

The SMACNA rectangular duct reinforcement tables are entered on the table for the pressure class, in the row for the duct dimension — the greater dimension of a duct determines the duct gauge for all four sides, reinforced or unreinforced. Each row then runs across a series of reinforcement spacing options from 10 ft down to 2 ft, and each cell gives a minimum duct gauge as a number and a minimum reinforcement grade as a letter. The manual states that 'any cell within a row is an acceptable selection' for that duct width, so gauge and reinforcement spacing are chosen together: closer reinforcement permits a lighter wall. At 1/2 in. w.g. a 24 in. duct is 22 gauge left unreinforced, or 26 gauge with grade C reinforcement out to 6 ft spacing and grade B at 5 ft and closer. Air velocity and run length drive friction loss and noise, not the wall gauge.

Q9. Answer: A

A square elbow without turning vanes creates significant turbulence, high pressure drop, and noise as air separates from the outer wall. Turning vanes (sheet metal blades inside the elbow) guide airflow in a smooth arc, reducing separation, pressure drop by up to 75%, and noise. Required by many specifications for critical systems.

Q10. Answer: C

When a cylinder is unrolled flat, its length equals the circumference (πd). The stretchout line represents this unrolled length; heights at each division come from the true-length view.

Q11. Answer: A

VAV boxes contain a motorised modulating damper controlled by the zone thermostat. They vary airflow from maximum to minimum (or zero) to maintain setpoint temperature.

Q12. Answer: B

A grooved seam is made by hooking two folded edges together and then driving the raised hook down so it locks. The tool for that is a hand groover, a hardened steel bar with a groove milled across its face. The groove is set over the raised lock and struck with a mallet: the tool drives the seam down while the groove itself keeps the lock from being crushed flat. Groovers come in sizes and the groove must match the seam width, because too narrow a groove mashes the lock and too wide a groove will not seat on it. The wrong answers are all real shop tools doing other jobs. A hand seamer has flat jaws. It folds an edge and flattens it, and on a Pittsburgh lock it does close the pocket over the inserted edge, but here the lock stands proud of the sheet and flat jaws have no groove to seat over it, so the seamer would crush the raised hook instead of driving it home. A hand crimper reduces the end of a round pipe so it slips inside the next section. A setting hammer struck directly on a raised grooved seam flattens the lock and can split light gauge metal; it is for dressing a seam already driven home, not for locking one.

Q13. Answer: C

The BC Occupational Health and Safety Regulation, Part 12, section 12.120 Flashback prevention, reads: "Suitable safety devices to prevent reverse gas flow and to arrest a flashback must be installed on each hose in an oxyfuel system, between the torch and the regulator." Two separate jobs are named in that sentence, and in the trade they are met by two devices, often combined in one body: a reverse-flow check valve, which stops gas from one hose backing into the other, and a flashback arrestor, which quenches a flame already travelling back up the hose before it reaches the regulator and the cylinder. Both belong in the length of hose between the regulator and the torch, whether they are screwed to the regulator outlet or fitted at the torch end. Protecting the fuel gas hose alone leaves the oxygen hose open, and it is reverse flow into either line that creates the trouble: fuel gas driven back into an oxygen hose, or oxygen into a fuel hose, makes a burnable mixture inside the hose itself, which is why the regulation says each hose rather than one of them. Fitting the device at the cylinder valve ahead of the regulator puts the regulator and the entire hose run on the unprotected side, so nothing that can actually burn is covered. The requirement is written for each hose in an oxyfuel system and carries no exemption for a portable cart outfit as against a piped or manifolded supply. The same Part of the regulation also requires the operator to check gas welding equipment for defects, leaks, oil and grease before it is used, and section 12.118 requires regulators, reducing valves and hoses to be used only for the gas they were designed for.

Q14. Answer: C

SMACNA classifies duct systems by static pressure: 1/2" WG, 1" WG, 2" WG, 3" WG, 4" WG, 6" WG, and 10" WG. The pressure class determines sheet metal gauge, seam types, reinforcement requirements, and joint spacing. Low-pressure residential systems are typically 1/2" or 1" WG.

Q15. Answer: B

Insulation on supply ducts in unconditioned spaces prevents condensation (cold ducts in humid attics), reduces heat gain from the attic, and improves system efficiency.

Q16. Answer: C

Plenum spaces contain energised conduit, communication cables, and the ceiling grid — contact with live wiring or stepping through the grid are key hazards requiring careful footing and awareness.

Q17. Answer: A

Flexible connections (canvas or neoprene) absorb AHU fan vibration, preventing noise and fatigue cracks in the rigid ductwork and building structure.

Q18. Answer: A

Heat tint is not just a colour. To build that oxide the surface gave up chromium, so under a heavy blue or black tint there is a chromium-depleted layer that is no longer as corrosion resistant as the parent metal. Wiping the colour off without taking that layer with it leaves the resistance still lost. Two routes remove both. Chemically, a pickling paste or bath strips the oxide and the depleted layer, and the surface is then passivated so a fresh chromium oxide film reforms. Mechanically, abrading cuts the layer away, using flap discs, wheels and brushes kept for stainless and never used on carbon steel. What matters either way is that nothing carbon steel touches the surface: a carbon steel wire brush, or a disc that has been run on mild steel, embeds iron particles that rust in service and stain the work, which is precisely the failure the cleaning was meant to prevent, so brushing and wiping makes the job worse rather than better. Reheating adds oxide rather than removing it. And the tint is not the passive film: the passive film is invisible, only a few atoms thick and self-repairing, whereas heat tint is a thick oxide sitting on depleted metal. The better answer is not to grow heavy tint in the first place - argon coverage, a gas lens, enough post-flow, and a back purge on the root side.

Q19. Answer: D

Heat tint is chromium oxide, formed when hot stainless is exposed to oxygen, and its colour is a rough measure of how much oxide grew and how much chromium was drawn out of the surface underneath to build it. A straw or light gold tint is minor and is normally accepted on non-critical HVAC work. Dark blue and grey mean heavy oxidation, and the chromium-depleted layer beneath that tint no longer carries the corrosion resistance of the parent metal, so at this colour the tint has to be taken off and the shielding put right. The cause is oxygen reaching the metal while it is still hot: too little argon flow, too great a distance from cup to work, a torch angle that draws air into the shielding stream, or moving the torch away before the post-flow has cooled the weld under gas. Mill scale is a different thing altogether. It is the oxide skin a hot rolling mill leaves over a whole sheet before any welding is done, dull and uniform across the surface, whereas what is described here appeared during the weld and is graded, running from straw at the outer edge through gold and purple to blue nearest the fusion line. Stainless duct stock is supplied cold rolled and pickled with a bright finish, so there is no rolling scale on it to find. The filler rod is not the cause either: composition governs how the deposited weld metal behaves and shows up as cracking or as poor corrosion behaviour in the bead itself, not as a graded colour band on the parent metal beside it. Prevention: correct flow rate, a gas lens, correct torch angle, and enough post-flow to cool the weld under shielding, with the root side back-purged where it must stay bright.

Q20. Answer: C

In most Canadian provinces (BC, Ontario, Alberta), fall protection (guardrail, safety net, or personal fall arrest system) is required when a worker is exposed to a fall of 3m (10 ft) or more.

Q21. Answer: D

A cone is rolled by giving the sheet more curvature at one edge than at the other, and on a three-roll slip roll that is done at the rear roller: one end of it is raised (or lowered) so the roller sits at an angle to the front pair. The edge of the sheet running through the closer end takes the tighter radius and becomes the small end of the cone, and the edge at the open end takes the larger radius. The two front rolls are the pinch pair. They grip and drive the sheet and their gap is set to the material thickness, so setting one of their ends tighter would only pinch and mark the sheet at that end rather than changing the curvature, which is set behind them. Feeding the sheet at an angle changes where the metal enters, not how hard it is curved, and it makes the cone spiral away from the operator instead of tapering. There is no tapered die: a slip roll forms by roller position alone, which is why the same machine rolls any diameter within its range without tooling changes.

Q22. Answer: C

Parallel line development is used for prisms (rectangular ducts) and cylinders (round ducts). The element lines are drawn parallel to the duct axis. On a prism they fall at the corners, so the stretch-out is laid off with the true width of each face; on a cylinder the circumference is divided into equal parts. Radial line development is for cones and pyramids, where the elements radiate from an apex. Triangulation is for transition pieces between different shapes.

Q23. Answer: A

SMACNA's text on sealing says it directly: contractors using slip and drive connection systems must control connector length and notch depth on rectangular duct ends, and failure to do so will compromise seal effectiveness. The reason is the corners. A drive cut long, or an end notched deeper than it needs to be, leaves an opening at the corner where the S-cleat, the drive and the two duct ends all meet, and that opening is the hardest place on the joint for a sealant to bridge. Liquid sealants are the type used on slip joints because they fill the small space between overlapping metal, but they are 30 to 60 percent volatile solvent and shrink considerably as they dry, so where the metal clearance exceeds 1/16 in (1.6 mm) several applications may be needed just to fill the void left behind. A corner gap opened by careless notching is far wider than that. Sealant brand and application temperature matter to the product's own performance but they are not what SMACNA names here, sheet gauge and pressure class govern the construction rather than the sealing of the corner, and rivet dimensions belong to a fastened lap rather than to a slip-and-drive joint, which is not riveted.

Q24. Answer: A

The grooved seam (or rolled groove seam) is used on round ductwork — particularly the spiral seam formed by a spiral pipe machine — and for connecting two curved edges. Both edges are formed into matching grooves and then rolled together to lock. It creates a smooth, mechanically tight seam without exposed sharp edges.

Q25. Answer: B

What decides this is medical treatment, not lost time. A nick dressed at the shop and entered in the first aid record goes no further, but a wound closed with sutures at a clinic is health care, and it engages two separate employer duties that run alongside each other. On reporting: section 150(1) of the Workers Compensation Act requires an employer to report to the Board, WorkSafeBC, within three days after its occurrence, every injury to a worker that is or is claimed to be an injury arising out of and in the course of the worker's employment. That duty does not wait to see whether a shift is lost, and it is the employer's own - whatever report a clinic or an attending physician makes on its own account, section 150(6) makes an employer who fails to report guilty of an offence. Section 150(7)(a) allows the Board to except a category of minor injuries from reporting by regulation, which is the space a first aid entry occupies, but the Act sets the investigation threshold at injury requiring medical treatment, and closing a wound with sutures is medical treatment. On investigating: section 69(1)(b) puts an incident that resulted in injury to a worker requiring medical treatment among those the employer must investigate. Section 71 requires the preliminary investigation immediately, with its report completed within 48 hours, and section 72 the full investigation, whose report goes to the Board within 30 days. Nothing in that turns on the injured worker asking for it. Other provinces impose their own versions of both duties, on their own bodies and their own time limits, which is why the scenario names a province.

Q26. Answer: A

The colour on the handles of aviation snips tells you the cut the blades are ground for, and yellow is the straight pattern: it takes a straight line and will also follow a large, gentle curve in either direction. Red cuts to the left, curving counterclockwise, and green cuts to the right, curving clockwise. Note what does not distinguish them: every aviation snip is a compound-action snip, which is why the same tools are sold as compound-leverage snips, so calling a snip compound action says nothing at all about which cut it makes. Hawk's-bill snips are a different tool. Their narrow curved blades are made for cutting small-radius circles and inside curves and cannot be held to a straight line.

Q27. Answer: A

The stretch-out, or girth, of a rectangular duct is the distance right around its cross-section: twice the width plus twice the depth, so $2 \times (400 + 250) = 1300$ mm. That figure covers the finished surface of the duct and nothing else. The Pittsburgh lock is made from metal that is not part of that finished perimeter. The pocket rolled on one edge and the plain folded edge that enters it are both extra metal, so their allowances are added to the girth rather than found inside it, and the same is true of any hem or flange added at the ends. Taking the lock allowance out of the girth runs the calculation backwards and leaves the section short by the depth of the lock, which cannot be made up once the sheet is cut. Halving the girth to 650 mm would be right only if the section were made from two L-shaped halves seamed at two corners, which is a different construction from the one the question describes. The Red Seal Occupational Standard puts the rule in the performance criteria for pattern development: the pattern points are connected, and allowances for seams and edges are added.

Q28. Answer: D

Tack welds are short welds that hold the parts in the position they were fitted up in, so the assembly cannot shift or pull out of square while the seam is welded. They are not the finished weld: they carry only enough strength to hold position, and the seam weld still has to be made. Tack quality matters, because a tack that cracks under the heat of the seam weld becomes a defect in the root of that weld. Good practice is to tack the ends first to hold the piece square, then add tacks between them to control distortion, spacing them to suit the gauge, the length of the joint and how much the work is moving. Tacking does nothing to clean the joint either: mill scale is taken off by grinding or wire brushing before the weld is struck.

Q29. Answer: B

A Pittsburgh lock former does not fold the pocket in one action. The edge passes through a train of roll sets, each bending it a little further than the last, so the pocket is formed progressively along the whole length as the panel travels through. That is why presentation matters. The panel lies flat on the machine table and the edge to be formed is fed straight into the first roll set and kept tight against the guide, so the same amount of metal enters every pass and the finished pocket is the same depth from one end of the panel to the other. Start the edge in at an angle and the amount of metal in the rolls changes as the panel goes through, which tapers the pocket and can jam or spring the roll set. Standing the panel on edge takes the flat off the table and leaves nothing keeping the panel travelling straight, so orientation is very much not arbitrary. And the rolls are driven one way only, from infeed to outfeed, so a panel cannot be entered from the outfeed side at all.

Q30. Answer: C

Triangulation sequence: 1) Draw plan and elevation views. 2) Divide the surface into triangles (diagonals across each face). 3) Find the true length of each line using a true length diagram (rotate each line to horizontal and measure). 4) Draw each triangle in sequence using the true lengths, building the flat pattern triangle by triangle.

Q31. Answer: B

Balancing dampers are manually adjusted during commissioning to set branch airflows to design values. They are locked in position after balancing is complete.

Q32. Answer: D

The metal that closes over a wire has to travel right around it, so the allowance is governed by the wire, not by the sheet. Sheet Metal Drafting, published in full and free to read at Wikisource, gives the shop rule: for covering a wire with metal, add an edge to the top of the pattern equal to two and a half times the diameter of the wire. It adds the qualification that matters on heavier work - this allowance must be increased for metal heavier than No. 23 gauge, because thicker metal cannot follow the wire tightly and takes a larger radius at the fold. Basing the allowance on sheet thickness picks the wrong variable: change to a heavier wire on the same sheet and the allowance changes outright, while changing to a slightly heavier sheet with the same wire changes it only a little. One wire diameter is not enough metal to reach around the wire at all, let alone to close back onto the panel. A fixed figure fails the moment the wire size changes, which is the whole point of tying the rule to the wire. Like every seam and edge allowance, the wired edge allowance is added to the true pattern after the pattern itself is drawn, and the edge is started in a bar folder and closed with the wiring rolls of a rotary machine.

Q33. Answer: C

A bar folder bends the edge of a sheet to a set width and angle in one stroke. The wing is adjustable, so the same machine will make a light break, a square flange or a flat hem, and because the width is set on the machine every panel comes off with an identical edge. The male half of a Pittsburgh lock is exactly that kind of edge: a single flange bent to about a right angle, which is entered into the pocket before the lip of the pocket is closed over it. The Pittsburgh lock former makes the other half of the same seam, the pocket, not the plain flange that goes into it. A slip roll former curls flat sheet into cylinders and cones between three rolls and puts no sharp fold in the metal at all, which is the whole point of it. A notcher only removes metal at the corners so that folded edges can meet without bunching; it forms nothing.

Q34. Answer: B

A box and pan brake, also called a finger brake, has a segmented upper beam. The fingers can be removed and rearranged so that the beam clears flanges already turned up on the workpiece, which is what lets the second, third and fourth sides of a box or pan be bent without fouling the sides already formed. Individual fingers are combined to make up whatever width the next bend needs. Rolling sheet into cylinders is the slip roll's job, cutting blanks to length is the squaring shear's, and a Pittsburgh pocket is rolled on a lock former.

Q35. Answer: C

LOTO (Lockout/Tagout) is mandatory before working near or on energised equipment. The sheet metal worker must de-energise, lock out, and verify zero energy state.

Q36. Answer: B

In parallel line development the plan view of the duct is divided into equal parts and those same divisions are then stepped off along the stretch-out line, so the circumference divisions and the stretch-out divisions are one and the same set and neither can determine the other. What sets the number is how closely the pattern has to follow the true curve. Each division produces one point where the mitre line is projected across onto the stretch-out, and the pattern outline is drawn by joining those points, so a coarse division cuts corners off the curve while a fine one traces it closely at the cost of layout time. Twelve or sixteen equal parts is the usual shop compromise. The Red Seal Occupational Standard states the criterion outright in sub-task B-6.02: the plan and elevation are divided into equal parts to achieve required accuracy. The number of gores decides how many separate patterns an elbow needs, not how finely each one is divided. Sheet gauge and bend allowance change the flat blank size and the seam and edge allowances, not the spacing of the division lines. Centreline radius divided by diameter is the elbow throat ratio, a fitting-selection figure with nothing to do with layout spacing.

Q37. Answer: B

The Pittsburgh lock (also called Pittsburgh seam) is the standard longitudinal closing seam for rectangular ducts. One edge is formed into a pocket (the "Pittsburgh pocket"), and the mating flat edge is inserted and then clinched (hammered) closed. It provides a strong, airtight closure along the full length of the duct.

Q38. Answer: D

Work it from the turn rather than from the piece count. Four pieces meet at three joints, and the fitting has to swing the duct through 90 degrees in all, so each joint turns it 30 degrees. A mitred joint is cut so the two mating faces lie flat against one another, which puts the cut plane on the bisector of the turn at that joint: each of the two pieces meeting there is cut 15 degrees off square. Every cut in the elbow is then identical, which is the point of laying it out this way, because one pattern serves the whole fitting and the pieces nest on the sheet. The pieces account for the turn as 15 plus 30 plus 30 plus 15: an end piece is cut on one end only and finishes square where it meets the straight duct, so it takes 15 degrees of the turn, while a middle piece is cut at both ends and takes 30. Thirty degrees is the turn the joint makes rather than the cut, and taking it for the cut forgets that a joint is shared between two pieces. Twenty-two and a half degrees divides the turn by the number of pieces instead of by the number of joints. Forty-five degrees is the cut for a two-piece 90 degree elbow, where one joint takes the whole turn. The BCcampus layout text works this same four-piece 90 degree elbow and reaches the same 15 degrees by its own arithmetic, quoted in the source.

Q39. Answer: A

ASHRAE 62.1 sets minimum outdoor air ventilation rates by occupancy type, expressed as a rate per person plus a rate per unit of floor area. That is the Ventilation Rate Procedure in Section 6.2: $V_{bz} = R_p \times P_z + R_a \times A_z$, where R_p (per person) and R_a (per unit floor area) are read from Table 6.2.2.1 in L/s or CFM. **It carries weight in Canada only because the Code adopts it: Sentence 6.3.1.1.(2) of Division B of the 2024 Ontario Building Code requires outdoor air to be supplied in accordance with Section 6.2 (Ventilation Rate Procedure), Section 6.3 (Indoor Air Quality Procedure) or Section 6.4 (Natural Ventilation Procedure) of ANSI/ASHRAE 62.1, and Sentence 6.3.1.1.(3) requires exhaust ventilation in accordance with Section 6.5.** In the 2012 OBC the same requirement sat at Sentence 6.2.2.1.(2), so older references and study notes will quote that number. Always use the edition your provincial code references, not the newest ASHRAE printing: the 2024 OBC and the National Building Code 2020 reference ASHRAE 62.1-2016, while the 2012 OBC referenced 62.1-2010. Because the 2016 edition dropped dwelling units from its scope, residential ventilation is covered by ASHRAE 62.2 or by Subsection 9.32.3 and CAN/CSA-F326, not by 62.1. The distractors belong to other documents entirely: duct velocity is a design and noise matter, refrigerant charge limits come from ASHRAE 15 and CSA B52, and flue gas CO₂ is a combustion analysis reading. None of the three appears in 62.1.

Q40. Answer: B

Transfer air grilles are typically installed high in the partition wall. Warm supply air accumulates near the ceiling and pressure differences drive transfer. Some codes allow door undercuts as an alternative.

Q41. Answer: A

An aviation snip has a small ramp on one jaw, just ahead of the pivot bolt. Cut with that jaw facing the waste and the ramp lifts the scrap up and over the pivot, while the piece being kept passes under the other jaw and stays flat and unpuckered. That is what lets the snip follow a curve without distorting the finished edge, and it is why the pattern is made in three hands, straight, left cut and right cut, so the ramped jaw can be put on the waste side whichever way the line runs. Driving the waste down under the lower blade is not what an aviation snip does, and it is the finished piece, not the handles, that the curl is being kept away from. Colour coding does tell you which way a snip will cut, but it does not make the waste side a matter of choice.

Q42. Answer: A

Correct soldering: heat the base metal (the seam), not the solder. When the seam is hot enough, touch the solder wire to the metal — it melts and wicks into the joint by capillary action.

Q43. Answer: B

Brazing: filler melts (liquidus >450°C/840°F) but base metal remains solid. Welding: base metal melts. The key distinction is whether the base metal is melted.

Q44. Answer: B

An eccentric reducer is an oblique cone. Its element lines still run to a single apex, because every cone has one, but the apex sits off to one side, so the elements are of unequal length. Each one must therefore be found in true length, by revolution or from a true-length diagram, before the pattern is struck. A right cone is the special case in which all the elements are equal, and that is what lets its pattern be swung with a single radius from one centre. Parallel line development suits shapes of constant cross-section, such as straight round pipe, not a taper. A concentric pattern cannot be rescued by trimming the small end either, because the offset changes the length of every element around the fitting.

Q45. Answer: D

AHU (Air Handling Unit) conditions and circulates air: it contains a fan, heating/cooling coils, filters, and dampers. Sheet metal ductwork connects to AHU supply and return.

Q46. Answer: C

A backdraft damper, also called a gravity damper, has light blades hung on a horizontal axis and held at or near the closed position by their own weight, sometimes with a small counterweight to set how easily they lift. The fan's airflow pushes them open in the direction of flow, and when the fan stops they fall shut, so air cannot travel back down the duct. Nothing drives them - no actuator, no wiring, no interlock with the fan - which is the condition the stem sets. A motorised isolation damper will also shut an idle duct, and on a large system it often does, but it needs power and a control signal, so it does not answer a question about an unpowered device. A fire damper sits open in normal service and closes only when its fusible link releases at its rated temperature; it protects a fire separation and takes no notice of which way the air is going. A manual volume control damper is set by hand during balancing and locked in that position, so it neither opens on forward flow nor closes against reverse flow. All four are damper types a sheet metal worker fabricates; what separates them here is how each one is operated.

Q47. Answer: B

Spot welding passes high current through copper electrode tips pressed against both sheet surfaces. Resistance heating at the contact point melts and fuses the sheets at discrete spots. Advantages: fast, no filler metal, minimal distortion, no flux. Limitations: only for lap joints, limited penetration, requires electrode access to both sides.

Q48. Answer: C

Cut-resistant gloves (ANSI/ISEA level A4+ or equivalent) protect against the sharp edges of cut sheet metal. Standard leather gloves provide limited cut resistance.

Q49. Answer: C

A rotary machine, also called a jenny or turning machine, works sheet between a pair of shaped rolls: swap the rolls and it turns edges, forms flanges, wires an edge, beads a panel for stiffness or swages a groove. Every one of those is a rolling operation. Punching a rivet hole removes material, which needs a punch press, an ironworker or a drill, so it is the one job on this list a rotary cannot do. Note that a Pittsburgh lock is also outside a rotary's range — it takes a dedicated Pittsburgh lock former — which is why it does not appear as a choice here.

Q50. Answer: A

A clogged/loaded filter has high resistance, causing elevated pressure drop across the filter section. This also reduces airflow. Filters must be replaced or cleaned.

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